----- Active space -----

PRESENTATION

- m/c : Asymptomatic.
- m/c symptom : Heavy menstrual bleeding (HMB).
- Others :
  - 2° dysmenorrhea.
  - Subfertility.
  - Pressure symptoms.
  - Pain.
  - Adverse pregnancy outcomes.

Cause of HMB :

- ↑ surface area of endometrium.
- ↑ vascularity of uterus : ↑Estrogen.
- Interference with uterine contractility.
- Ulceration of Submucosal fibroid.
- m/c fibroid causing HMB : Submucosal fibroid.

Cause of Subfertility :

- Impaired gamete transport d/t impaired → Tubal motility. / Uterine contractility.
- Interference with implantation d/t distorted uterine cavity.
- Displacement of cervix away from vaginal pool of semen.
- m/c fibroid causing subfertility : Submucosal fibroid.

Pain :

- Usually painless
- If painful : Pelvic pressure/pain.
- Acute pain abdomen with fibroids → Torsion / Red degeneration of fibroid

Pressure symptoms :

- Urinary : Cervical fibroid.
- Rectal : Post cervical fibroid (rare).

Endocrine symptoms :

- Polycythaemia.
- Hypercalcemia.
- ↑ PRL levels.

----- Active space -----

Pseudo meig's syndrome :

- Pedunculated fibroid
  +
- Right sided pleural effusion
  +
- Ascites

Mimics → Malignant ovarian tumor

myomectomy ↓

Spontaneous resolution.

Note :

Not seen in fibroid : Amenorrhoea.

Pregnancy Complications :

- Pre term labor.
- IUGR.
- Abruptio placentae.
- Malpresentation.
- PPH (Post partum hemorrhage).
- Subinvolution.

DEGENERATIONS OF FIBROID

Physiology of degeneration :

- Most vascular part : Periphery.
- Least vascular part : Central.

↓

Degeneration begins from centre of fibroid.

Hyaline degeneration :

- m/c degeneration.
- Homogenous appearance of fibroid.
- Whorled appearance : Lost.

Calcareous degeneration :

- m/c seen : Subserosal fibroid
- Begins from periphery
- X-Ray : Popcorn calcification / Womb stone appearance.

Sarcomatous Degeneration :

- Least common degeneration (<0.5%)
- m/c seen in : Intramural fibroid.

----- Active space -----

Clinically :
Peri/post menopausal female.
↓
- Sudden ↑ in size of fibroid
- Becomes painful.

HPE (Histopathological Examination) :
- Becomes soft.
- Loose pseudocapsule.
- ≥10 mitoses/ HPF.

Note :
- < 5 mitoses/HPF : Benign fibroid
- 5-10 mitoses/HPF : STUMP
- Smooth muscle tumor of uncertain malignant potential.

Red degeneration :
- Specific to pregnancy.
- m/c seen : 2nd half of pregnancy.

Pathogenesis :
Thrombosis of blood vessels supplying fibroid.
↓
Aseptic necrosis of fibroid + Peritoneal irritation.

Presentation :
Pregnant female with c/o : Pain abdomen + nausea/vomiting ± fever.

Investigations :
- ↑ WBC.
- ↑ ESR.

Management :
- Conservative Mx
  - Analgesics.
  - Antiemetics.
  - IV fluids.
- NO :
  - Antibiotics.
  - Termination of pregnancy.
  - Myoectomy.

Note :
- 'Red' degeneration :
  - D/t : Thrombosed vessels —release→ Blood pigments —diffuse→ Salman pink fibroid.
  - Fishy odor.

----- Active space -----

INVESTIGATIONS

- IOC : TVS (Trans Vaginal Sonography).
- Submucous fibroid :
  - Saline infusion sonography (SIS) > TVS.
  - Hysteroscopy.
- MRI :
  - Rarely needed for Dx.
  - Maybe done Pre-operatively.

Management of Fibroid

00:25:40

1. ASYMPTOMATIC

Observation,

Except

Submucous fibroid + C/o infertility

- Type 0, 1 : Hysteroscopic myomectomy.
- Type 2 onwards : Laparoscopic myomectomy.

B/L cornual intramural fibroid + C/o infertility

Laproscopic myomectomy

Note :

- Generally,
  - Submucous + symptomatic fibroid → Sx Mx.
  - Intramural
  - Subserosal

  } + Syptomatic → medical Mx
- Exception : B/L cornual intramural fibroid + infertility → Sx Mx.

----- Active space ----- **SYMPTOMATIC**

i. C/o HMB/AUB :

USG
- Intramural/subserosal fibroid → Medical mx :
  1. 1st line drugs : ↓ Blood loss, but not size of fibroid
     - Tranexamic acid.
     - OCPs.
     - Progesterone (Oral/mirena).

     Fail ↓
  2. 2nd line drugs : ↓ Blood loss + ↓ Size of fibroid

     Principle : Fibroid E + P dependent —↓E or ↓P→ ↓size
     - ↓E :
       - GnRH agonist (continuous). } FDA Approved
       - GnRH antagonist. }
       - Letrozole
       - Danazol (S/E : Hirsutism)
       - Gestrinone
     - ↓P :
       - Mifepristone (RU 486).
       - Ulipristal.

     Fail ↓
  3. Interventions :

     Done only if : Family complete.

     a. Uterine artery Embolization (UAE) —Via→ C/L femoral artery using Polyvinyl alcohol /Gel form
     - Contraindications :
       - Absolute :
         - Patient wants to conceive
         - Active PID
         - Malignant fibroid.
       - Relative : Post menopausal

     b. HIFU (High Intensity Focused USG)

     c. MRGFUS (Magnetic Resonance Guided Focused Ultrasound).
     - Principle : High intensity USG waves —focused on fibroid→ Coagulative necrosis of fibroid

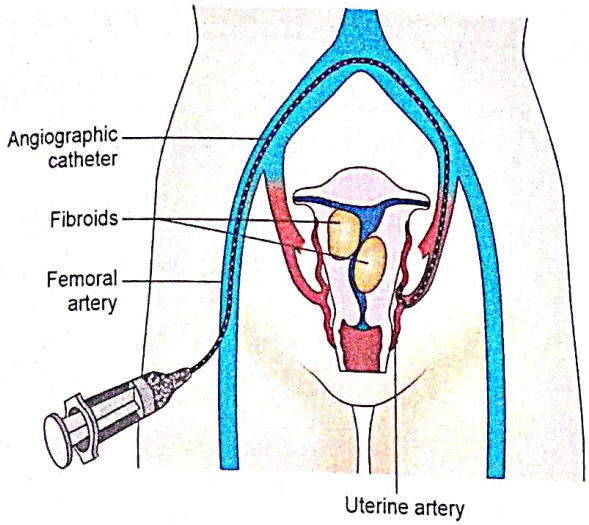
     Fail ↓
  4. Surgical

Submucous fibroid
- Type 0/1 → Hysteroscopic myomectomy
- Type 2/any type with si ≥ 5cms → Laproscopic myomecton

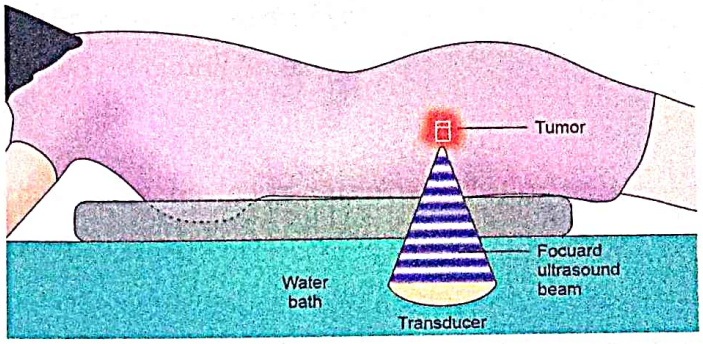
4. Surgical :

- Family complete : Hysterectomy
- Family not complete : Myomectomy

----- Active space -----



Uterine artery embolisation



HIFU

ii. C/o Pelvic pressure symptoms :

Seen : Intramural/Subserosal fibroid

Cause : Large size of fibroid.

Mx : Drugs that ↓size of fibroid.

- Continuous GnRH agonist.
- GnRH antagonist.

iii. C/o Infertility :

Seen with : Submucosal fibroid.

Mx : Surgery.

- Type 0 & 1 : Hysteroscopic myomectomy.
- Type 2 onwards : Laparoscopic myomectomy.
- B/L cornual intramural fibroid (rare) : Laparoscopic myomectomy.

MYOMECTOMY

Removal of fibroid leaving uterus behind.

Indications :

- C/o infertility,
- Family not complete.

Routes :

- Type 0/1, Submucosal fibroid : Hysteroscopic.
- Type 2/any type ≥ 5 cms : Laparoscopic.

----- Active space -----

Pre-myometomy :

- C/o HMB : Endometrial Biopsy to rule out hyperplasia.
- C/o infertility : Rule out other causes of infertility (male partner semen analysis).
- ↑ Blood loss during Sx :
  - Ix hemoglobin pre-op.
  - Arrange for blood.
  - Perform Sx in pre-ovulatory phase.
  - ↓ size of fibroid by giving GnRH.

During Sx :

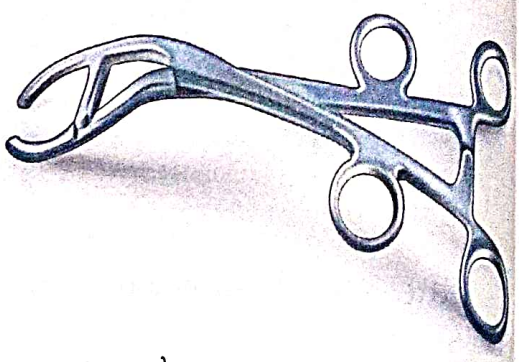
- Administer hypotensive anaesthesia.
- Give local vasopressin injection.
- Use clamps

1. Boney's myomectomy clamp :

Uses:

a. ↓ blood loss during Sx

b. Clamp B/L uterine artery (Including round ligament)
   - Clamp x 20 mins.
   - Release x 10 mins.

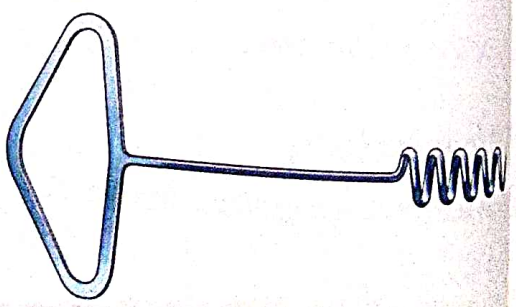


Boney's myomectomy clamp

2. Sponge holding forceps : To clamp ovarian artery.

- Myoma screw.

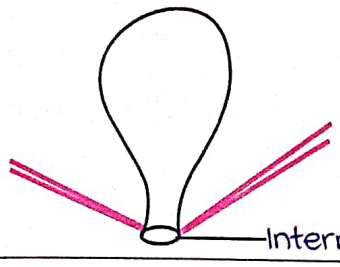
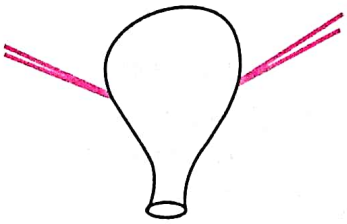
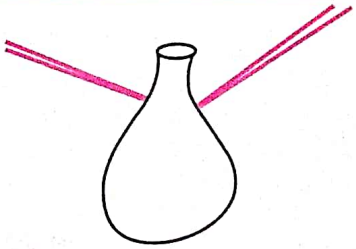
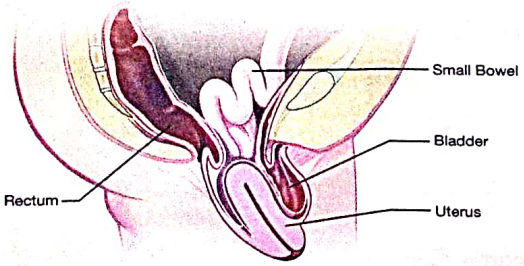
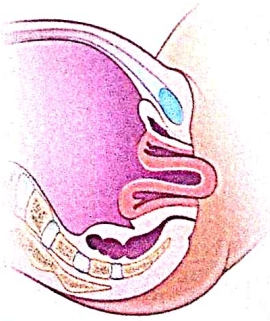
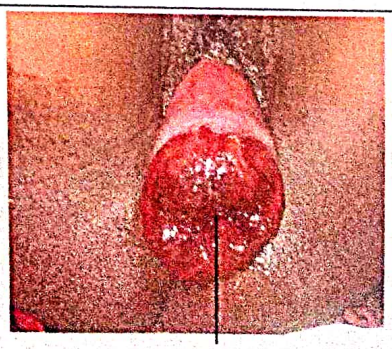
Use : Hold fibroid during Sx.



Myoma screw

Recurrence rate : 5-10%.

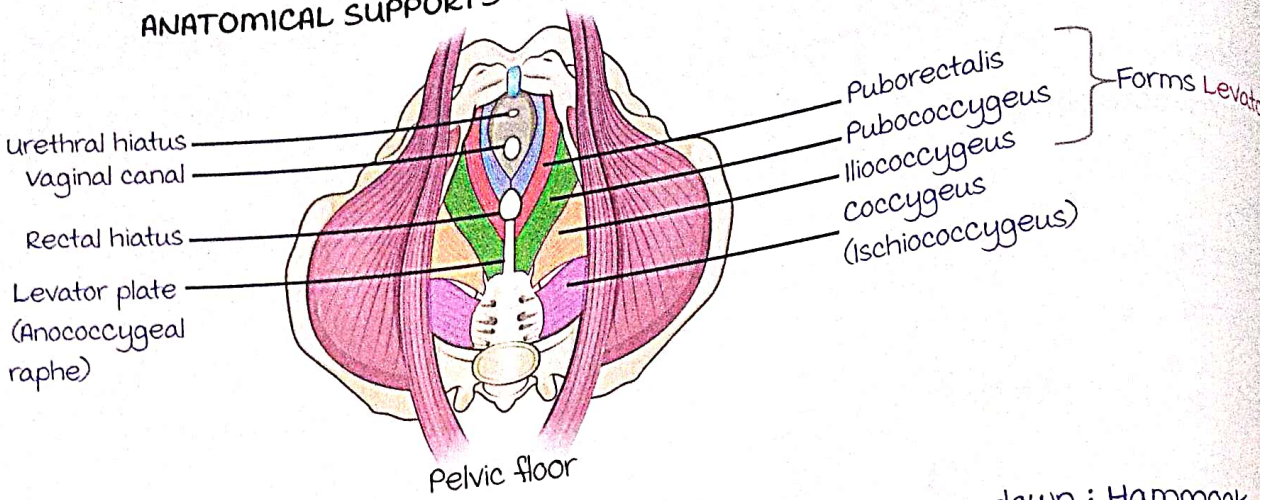
Continued menorrhagia post Sx : 10-15%.

Normal	Prolapse	Inversion
		
Feature	Os is visible.	Fundus protrudes out.
Cause	D/t loss of support.	Iatrogenic complication of labour.
Diagramatic representation		
Appearance		

----- Active space -----

Supports of Uterus

ANATOMICAL SUPPORTS

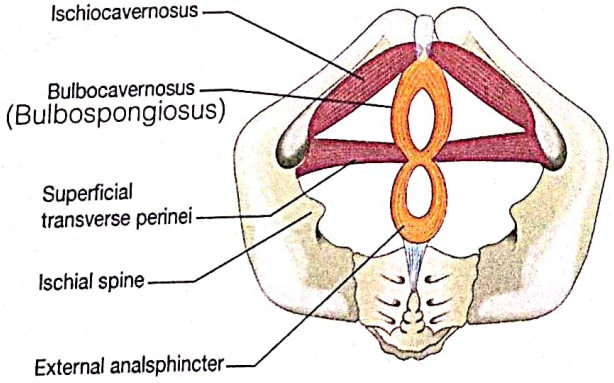


Pelvic floor

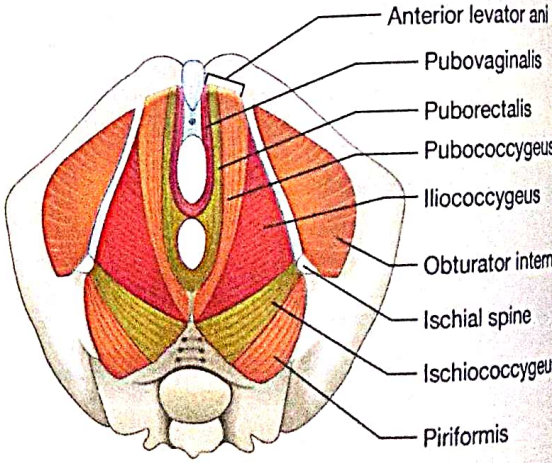
Pelvic diaphragm :

- Levator Ani + coccygeus (Ischiococcygeus).
- Voluntary muscle.
- State of muscle :
  - Contracted → Defecation
  - Relaxed → Urination, Child birth
- Hangs down : Hammock.
- Insertion : Levator plate.

Pelvic floor muscles :



superficial



Deep

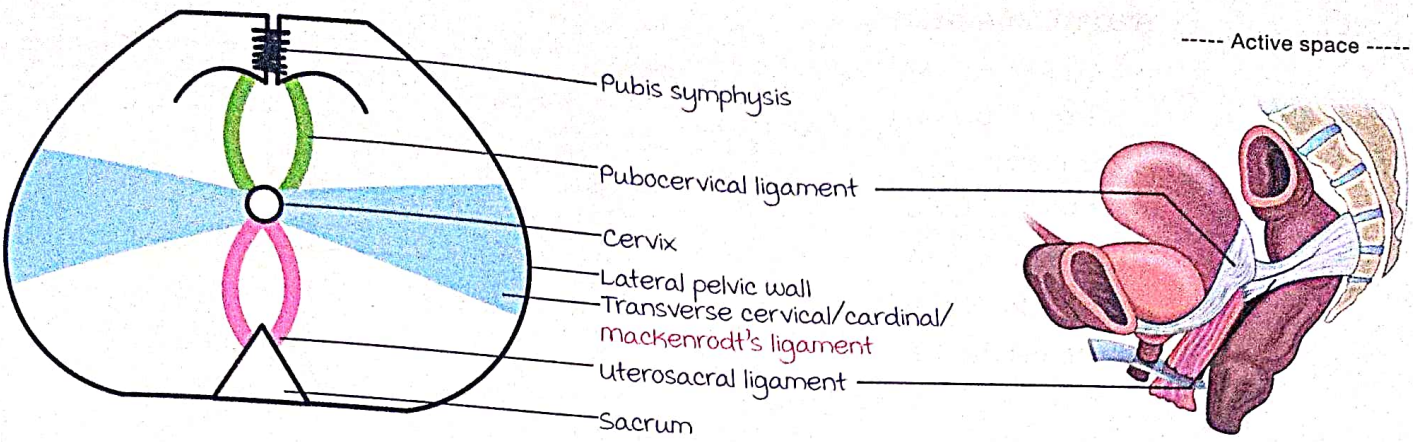
Muscles which do not support uterus :

1. Ischiocavernosus.
2. Obturator internus.
3. Piriformis.
4. Ischiococcygeus.
   - Lateral muscles.
   - Do not attach to perineal body.
   - Not cut during episiotomy.

Muscles which support uterus :

1. Levator ani muscle (most important).
2. Deep transverse perinei.
3. Superficial transverse perinei.
4. Bulbospongiosus.
5. External urethral sphincter.
   - Cut during episiotomy.

Ligaments :



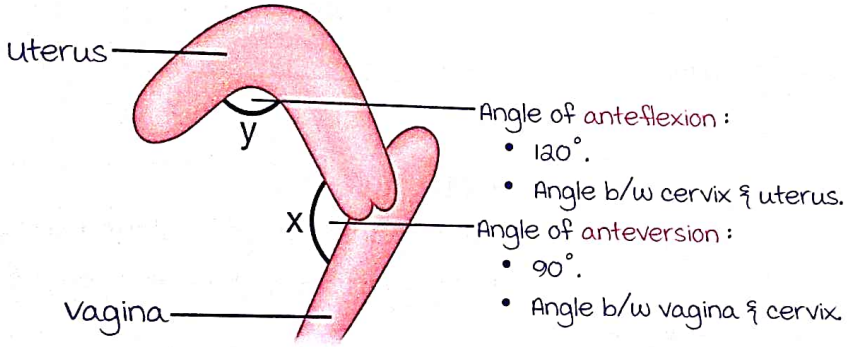
Ligaments which support uterus :

- 1° Supports of uterus : muscles + ligaments.

1. Pubocervical ligament.
2. Cardinal ligament (Strongest, most important).
3. Uterosacral ligament (and strongest).

} Tri-radiate ligament

MECHANIC SUPPORT



APPLIED ANATOMY

1. First step of prolapse : Retroversion.
2. Retroversion seen in :

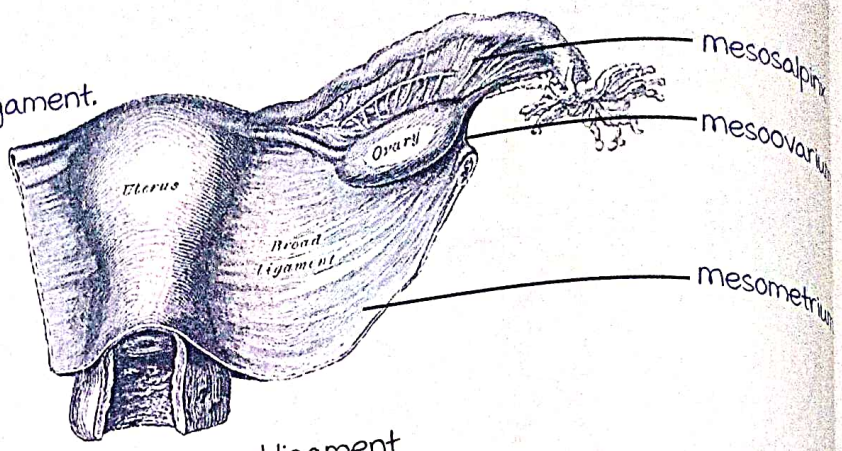
	Prolapse	Endometriosis
Corrected manually	⊕	⊖ : Fixed retroverted uterus

3. Ligaments maintaining AV position : Round ligament (RL) + uterosacral ligament (US) > RL > US.
4. Ligaments preventing retroversion : RL + US > US > RL.
5. Round ligament : Secondary support of uterus.
6. Ligament that does not support uterus : Broad ligament.
7. Most important support of uterus : Levator ani.
8. Damage to puborectalis —leads to→ pelvic organ prolapse.

----- Active space -----

BROAD LIGAMENT

1. Misnomer : Not a ligament.
2. Peritoneal fold.
3. 3 parts:
   - mesosalpinx.
   - mesovarium.
   - mesometrium.
4. Contents of BL:
   - 1 Tube : Fallopian tube.
   - 2 Ligaments :
     - Round ligament.
     - Ovarian ligament.
   - 2 arteries :
     - Uterine A.
     - Ovarian A.



Broad ligament

- 2 Nerve plexus :
  - Uterine plexus.
  - Ovarian plexus.
- 3 Embryological remnants :
  - Epoophoron.
  - Paroophoron.
  - Gartner's duct.

Risk Factors & Classification

00:40:23

RISK FACTORS

- m/c in : Elderly multiparous female.
- Birth trauma.
- Multiparity.
- Menopause d/t estrogen deficiency.
- Instrumental delivery.
- ↑Intra-abdominal pressure.
  - chronic constipation.
  - chronic cough.
  - long standing ascites.
- Obesity.
- Prolonged labour/obstructed labour.
- Collagen abnormalities : Ehler Danlos syndrome.
  - prolapse in young, multiparous female → Congenital prolapse.
- Hysterectomy : ↑risk of vault prolapse & enterocele.

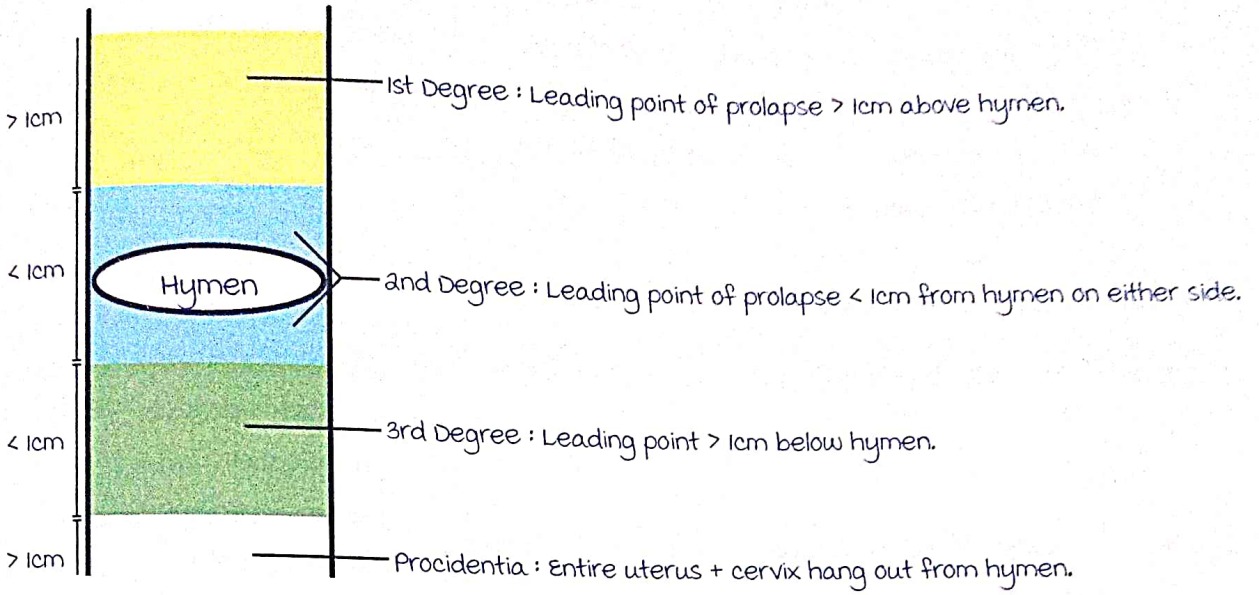
CLASSIFICATION

Classification system	POP Q	Show	Baden Walker
Reference point	Hymen	Introitus	Hymen

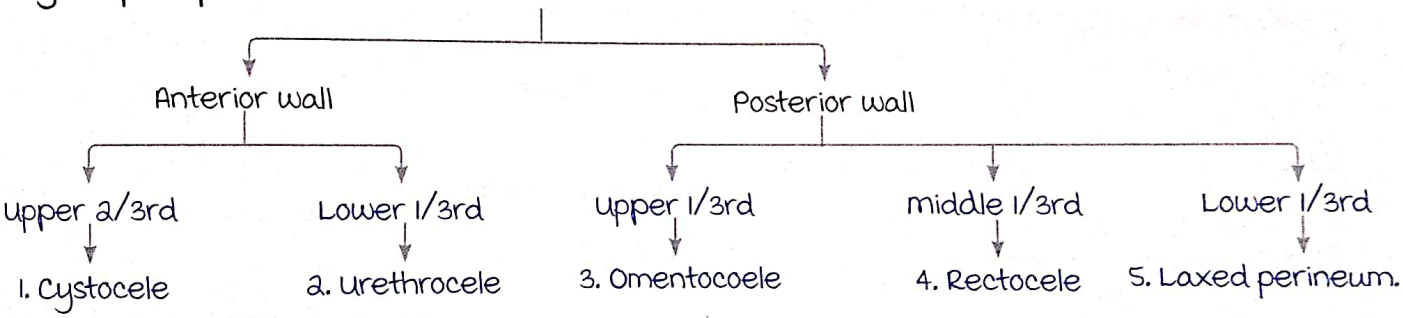
POP Q classification :

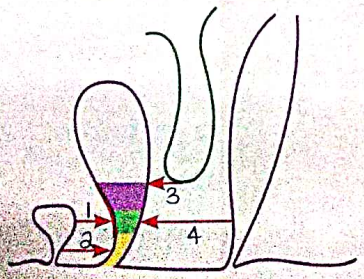
----- Active space -----

- Pelvic organ prolapse quantification classification.
- Reference point :

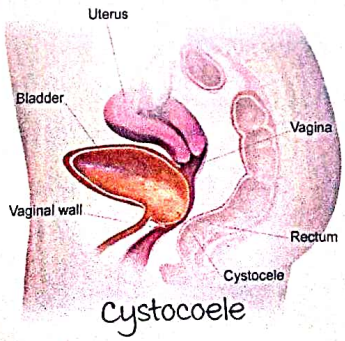


Vaginal prolapse Classification :

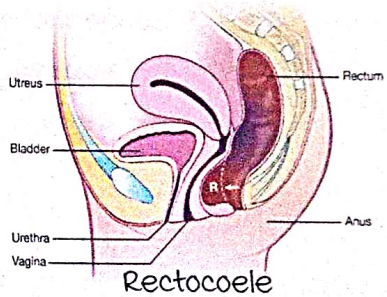




Vaginal prolapses



Cystocoele



Rectocoele

Enterocoele

Rectocoele

----- Active space -----

Enterocoele vs rectocoele :

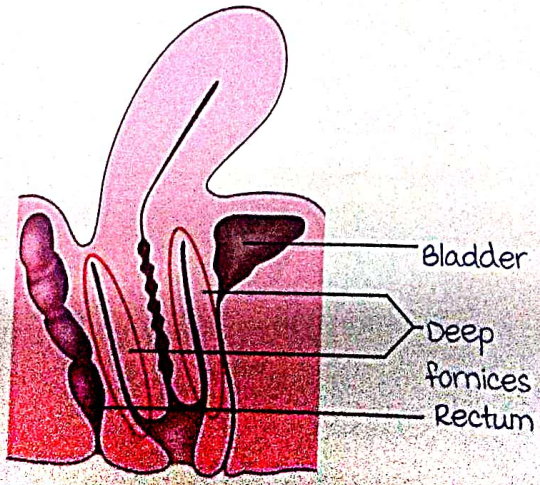
	Enterocoele	Rectocoele
P/V Examination	• Bulge at upper 1/3rd of posterior vaginal wall. • Cough impulse +.	• Bulge at lower 2/3rd of posterior vaginal wall. • Cough impulse −.
P/R Examination	Bulge at tip of finger.	Bulge : Pulp of finger.
Transillumination	+ve	−ve

DE LANCY'S LEVELS OF SUPPORT OF VAGINA

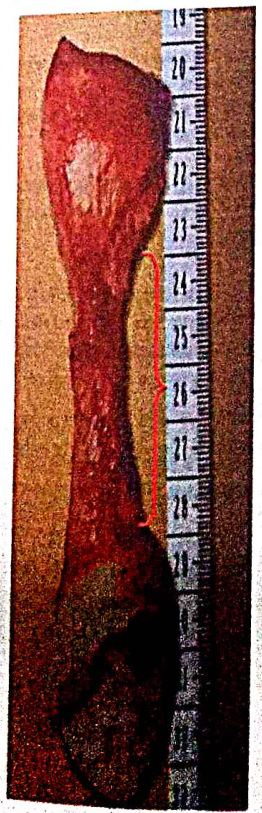
Levels	Part of vagina supported	Supports	Defects
Level 1	Upper 1/3rd of vagina	CL + US ligament.	1. Uterocervical prolaps 2. Apical prolapse. 3. Vault prolapse. 4. Enterocele. 5. Supravaginal elongation of cervix.
Level 2	middle 1/3rd of vagina	Arcus tendinous fascia.	1. Cystocele. 2. Rectocele.
Level 3	Lower 1/3rd of vagina	• Perineal body + muscles. • Urogenital diaphragm muscles.	1. Urethrocele. 2. Lax perineum. 3. Rectocele ±.

Note :

- Vault : part of vagina left after hysterectomy.
- Congenital elongation of cervix :
  - Vaginal part is elongated.
  - Deep fornix.



Congenital cervical elongation



Supravaginal elongation of cervix

PROLAPSE : PART 2

----- Active space -----

Symptoms & Management

00:00:08

SYMPTOMS

- Feeling of vaginal fullness.
- Something coming out of vagina.
- Low backache : D/t stretching of uterosacral ligaments.
- Urinary symptom : Cystocele/urethrocele.
  - Incomplete voiding of urine.
  - ↑ risk of UTI.
- Stress urinary incontinence d/t descent of bladder neck.
- Bowel symptoms : Rectocele/enterocele.
- Decubitus ulcer.

Decubitus ulcer :

Pathology :

- Venous congestion (D/t hampered venous return).
- Not d/t friction.

Decubitus ulcer

Management :

- Acriflavine (Antiseptic) + glycerine (Hygroscopic agent) packing.
  - Heals in 10-15 days.
- Betadine.
- Kegel's exercise : Pelvic floor muscle strengthening.
  - During pregnancy & after child birth.
  - Alternate contraction & relaxation of pelvic floor x 10 cycles x 3 times/day.

  Best prevention /Rx of 1st degree prolapse.

MANAGEMENT

Vaginal prolapse :

Mx independent of age/parity.

Sx for cystocele/urethrocele :

Anterior colporrhaphy : Excision of prolapsed vaginal wall —Repair retained walls→ Interrupted sutures

----- Active space -----

Sx for enterocele :

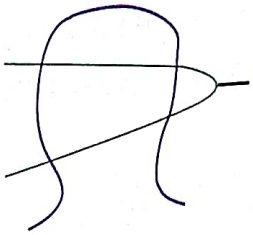
- McCall culdoplasty : Also done prophylactically post hysterectomy to prevent enterocele.
- Outdated Sx :
  - Moschcowitz repair.
  - Halban repair.

Sx for rectocele/ laxed perineum : Posterior colpoperineorrhaphy.

Uterine Prolapse :

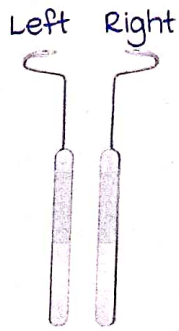
Age dependent mx.

Condition	management
1. Uterocervical prolapse in a female who wants future child bearing • Pre-requisite : – Uterocervical length : Normal	Sling Sx/Cervicopexy



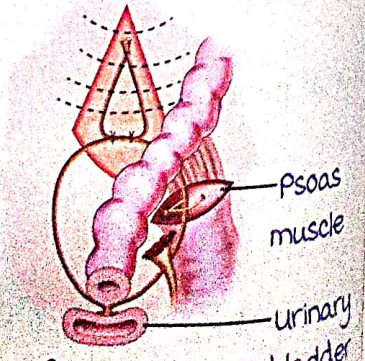
- Sling made of : mersilene tape.
- Sling can be :

a) Anterior : Purandare sling
- One side : Isthmus of uterus.
- Other side : Rectus sheath.
- Dynamic sling.
- Complications : Less.
- success Rate : Low.



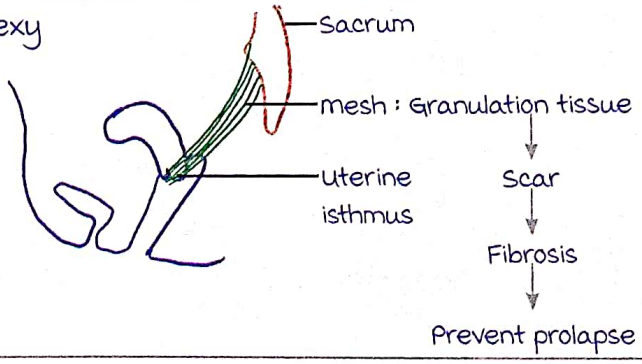
Shirodkar needle

b) Posterior : Shirodkar sling
- One side : Isthmus of uterus.
- Other side : Ligaments attached to sacral promontory.
- Static sling.
- Complications :
  - i. Seen on left side :
    1. Intestinal obstruction (Sigmoid colon).
    2. Mesenteric injury.
    3. Injury to genitofemoral nerve.
  - ii. Avoiding sigmoid colon obstruction : Hook made from psoas muscle.
- Success rate : High.

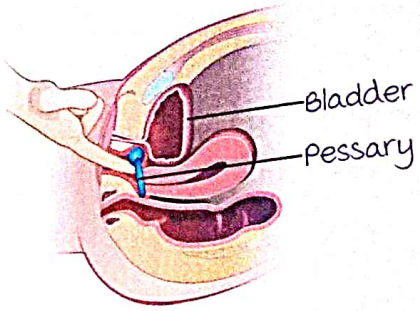


Psoas hook

----- Active space -----

Condition	management
	c) Composite Sling/Virkud sling • Left side : Purandare sling Sx. • Right side : Shirodkar's sling Sx. d) Khanna sling : Anterior sling surgery. • One side : Isthmus of uterus. • Other side : ASIS
2. Prolapse in a reproductive age female who doesn't need future child bearing Note : To prevent cervical stenosis	• Surgery : Fothergill repair/manchester repair. • Steps : 1. Cervical amputation. 2. Plication of cardinal ligament. • Complications of Fothergill Surgery. - Cervical stenosis. - Fibrosis. - Cervical incompetence. • Shirodkar modification of Fothergill Sx. - Only plication of cardinal ligament. - Can be done in ♀s needing future child bearing. Prevention of cervical stenosis : Prophylactic D & C + dilatation of cervix.
3. Nulliparous prolapse/prolapse in ♀ who want to retain uterus	Sacrohysteropexy 
Note: • Surgical options in ♀ wanting future child bearing : Sacrohysteropexy > Sling Sx > Shirodkar modification of Fothergill repair.	
4. Prolapse in menopausal femal	• Vaginal hysterectomy.
5. Prolapse in 60-65 year female, with some C/I to Sx (Diabetes/HTN)	• Lefort's colpocleisis : - Stitch anterior & posterior wall of vagina. - ↓ LA (Local anaesthesia). - Pre requisite : Rule out endometrial & cervical cancers. - Not done in : Reproductive age ♀s. - Also done in : mx of vault prolapse.

----- Active space -----

Condition	Management
6. Prolapse in old ♀ (> 65yrs) with comorbidities and unfit for any surgery	• Ring pessary : – Space occupying device. – Temporary cure. – Changed every 3–6 months.
7. Prolapse during pregnancy and prolapse during puerperium	• Ring/Gell horn/Donut pessary. • ≈ 16–18 weeks of pregnancy : – Remove pessary (Uterus : Abdominal organ → Automatic correction of prolapse 

Vault Prolapse

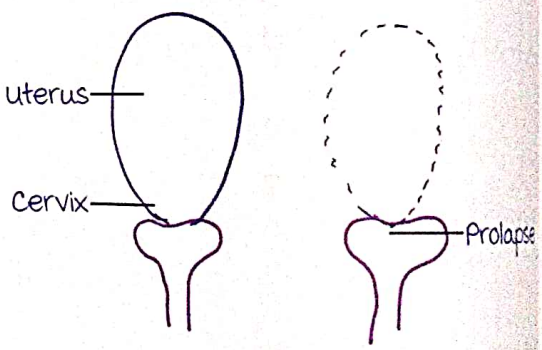
00:40:50

Prolapse of vaginal stump after hysterectomy.

PREVENTION

Uterosacral suspension :

- Done during vaginal hysterectomy.
- Also for Rx.



MANAGEMENT

1. Vaginal Sx :
   - Uterosacral suspension.
   - Sacrospinous fixation.
   - Lefort's colpocleisis.
2. Abdominal Sx :
   - Sacral colpopexy.

Note :

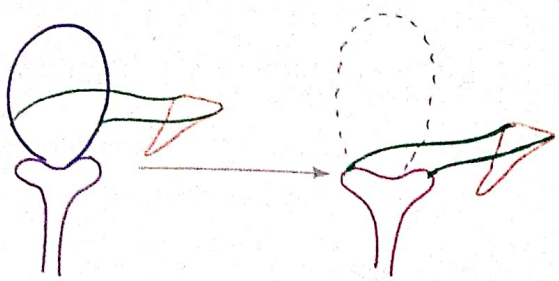
Burch colposuspension :

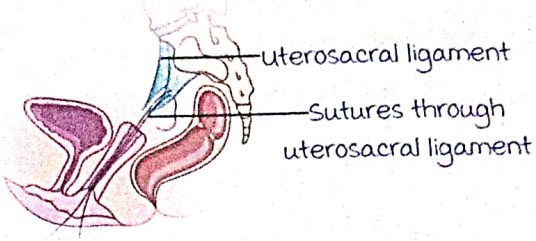
- Mx of Stress Urinary Incontinence (SUI).
- Not a Mx of vault prolapse.

----- Active space -----

Uterosacral suspension :

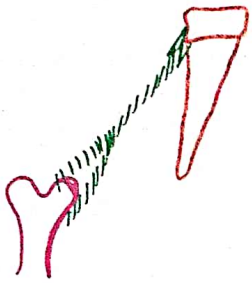
- Easy to perform.

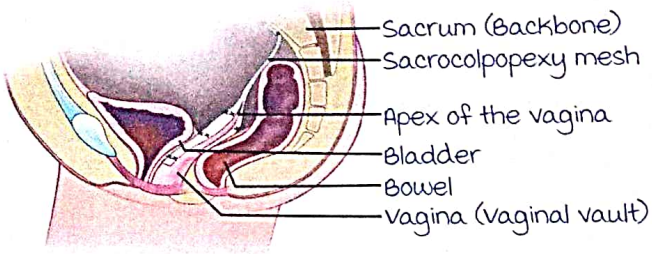




Sacral colpopexy:

- Best mx for vault prolapse.
- Difficult to perform.
- mesh placed between sacrum & vaginal vault.





Clinical Concepts of Prolapse

00:50:22

Examination :

Procidentia

Put prolapsed part in normal anatomical position
↓ without emptying bladder
Check for SUI
↓ Empty bladder

P/V examination :

- Size of uterus.
- Position (Retroversion).
- Check : Uterocervical descent/only vaginal descent.
- Assess degree of prolapse : Ask patient to strain/perform valsalva.
- Reduce prolapse : To normal position
  ↓ measure
  Total uterovaginal length

P/V examination

----- Active space -----

Management :

- Cystocele + Rectocele
  - Anterior colporrhaphy
  - Posterior colporrhaphy
  - → Pelvic floor repair
- Old ♀ + 3° uterine prolapse + Cystocele + Rectocele
  - Vaginal hysterectomy
  - Pelvic floor repair
  - → Ward Mayo's hysterectomy
    + McCall culdoplasty (Prevent enterocele)
    + Uterosacral suspension (Prevent vault prolapse)

NORMAL SEXUAL DEVELOPMENT

----- Active space ----

Development of Internal Genitalia

00:01:09

Development of Gonads :

- Gonads are bipotential.
- Derived from genital ridge at 5-6 wks of intrauterine life.
- Based on SRY gene develop into → Male gonads : Testis.
  → Female gonads: Ovaries.

SRY :

- Sex related gene on Y chromosome.
- AKA Testis determining factor.
- Location : Distal end of short arm of Y chromosome.
- Sex determination : SRY gene → Present → male child.
  → Absent → Female child.
- Micronutrient necessary : Zinc.

Intrauterine period :

Presence of 2 pairs of genital ducts

	Mullerian duct :	Wolffian duct :
AKA :	Paramesonephric duct	mesonephric duct
Derived from :	Outgrowth of coelomic epithelium	Intermediate mesoderm

At 9 wks : One pair of ducts → Internal genital organs

At 10 wks : Other pair of ducts → Regresses & leaves remnants.

Internal organs :

Males	Females
Seminal vesicles	Fallopian tubes
Epididymis	Uterus
Ejaculatory duct	Cervix
Vas deferens	Vagina (upper part)

Methods of Sex Determination & Applied Aspects :

methods	Applied aspects
Appearance of external genitalia (m/c)	Ambiguous genitalia : Difficulty in determining sex of a child by external genitalia.
Karyotyping : Best method. Y chromosome → Present : male. → Absent : female.	• Turner syndrome → 45,XO → Female. • Klinefelter syndrome → 47,XXY → male.
No of Barr bodies (No. of X chromosomes - 1)	Not effective in all cases (Eg. Turner syndrome, Klinefelter syndrome)

----- Active space -----

NORMAL SEXUAL DEVELOPMENT IN MALES

- Most important gene for Formation of testis : SRY > SOX-9.
- SRY gene —Activates→ SOX-9 → Formation of testis at 6 – 7 weeks.
- Testis : Hormonally active.
- 2 Types of cells :
  - Sertoli cells.
  - Leydig cells.

Sertoli cells :

FSH ⟵ (−) Inhibin B

FSH → Sertoli cells

- Contain receptors for SRY gene
- AKA sustentacular cells

Anti Mullerian Hormone (AMH)/mullerian Inhibiting Substance (MIS) :

- Time of synthesis → In males : Intrauterine life.
  → In females : At puberty from pre-antral follicles of ovary
- Functions :
  - Abdominal descent of testis.
  - Inhibit mullerian duct & regresses → Remnants :
    a. Prostatic utricle.
    b. Appendix of testis.
    c. Hydatid of morgagni.

Leydig cells :

LH → Leydig cells → Outside the blood-testis barrier

Testosterone :

- Synthesised from 8 weeks of intrauterine life.
- Functions :
  - Promote growth of Wolffian duct → male internal genital organs develop
  - Scrotal descent of testis
  - Negative feedback on GnRH → ↓ LH, FSH
  - Forms dihydrotestosterone by action of 5α-reductase → masculinization of external genitalia.

Note :

- 1st feature of testis development on USG : Appearance of testicular cords.
- Gubernaculum/Processus vaginalis → Causes descent of testis
  → Homologous structure in ♀ : Round lig.
- hCG (1st stimulus) : Stimulates testosterone production.
- Testosterone & Dihydrotestosterone : Same receptor.

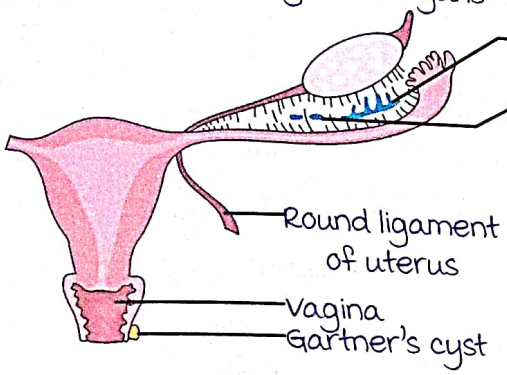
NORMAL SEXUAL DEVELOPMENT IN FEMALES

----- Active space -----

46 XX → Absent SRY gene

Formation of ovary : Hormonally silent in intrauterine period

No sertoli cells → Absent AMH → Growth of mullerian duct → Formation of internal genital organs



No leydig cells → No testosterone

Wolffian duct regresses : Remnants
- Epoophoron
- Paroophoron
- Gartner's duct

No testosterone → No dihydrotestosterone → No masculinisation of external genitalia → External appearance of female genitalia

Remnants of wolffian duct :

	Remnant of	Location
Epoophoron	Cranial end of mesonephric tubule	Leaves of broad ligament
Paroophoron	Caudal end of mesonephric tubule	
Gartner's duct	Mesonephric duct	Anterolateral wall of vagina

Development of ovary :

- Both X chromosomes + : Complete.
- Turner syndrome (XO) : Incomplete → Streak gonads.

Derivatives :

Part	Derived from
Upper vagina	Mullerian duct
Lower vagina	Sinovaginal bulb of Urogenital sinus
Ovary	Genital ridge

Miscellaneous points :

- No external stimulus required for development of female external genitalia.
- Appearance of → Testes : 6–7 weeks. / Ovaries : 7–8 weeks.
- Differentiation of female & male : 10 weeks.
- External genitalia development : 14 weeks
- MIS/AMH : Acts on ipsilateral mullerian duct → Unilateral development → True hermaphroditism/Ovotestis

----- Active space -----

Role of estrogen :

Growth of uterus at puberty.

Applied aspect :

- Turner's syndrome : Streak ovaries
- Gonadal dysgenesis
- Kallman syndrome : Hypothalamic failure

↓ GnRH → ↓ LH, FSH.

↓ Estrogen → Hypoplastic/Infantile uterus (Poorly developed uterus)

00:37:47

Development of External Genitalia

Depends on intrauterine testosterone.

Homologous Derivatives of male & female genitalia in intrauterine life :

	Testosterone +	Testosterone −
Genital tubercle	Penis	Clitoris
Labioscrotal swelling	Scrotum	Labia majora
Genital fold	Penile urethra	Labia minora

Source of testosterone :

In males :

- Testis
- Adrenal gland

In females :

- Adrenal glands
- Androgen $\xrightarrow{\text{Aromatase}}$ Estrogen

External genitalia depending on level of testosterone :

	XY individual	XX individual
Absent	Female	Female
Normal	Male	–
Low level	Ambiguous	Ambiguous

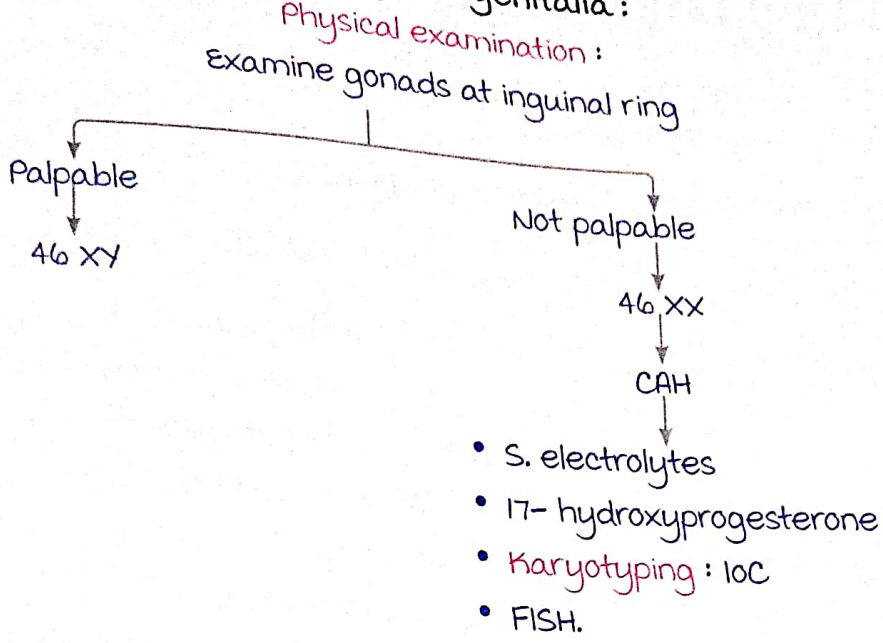
FEATURES DEFINING AMBIGUOUS GENITALIA

- Undetermined sex of the external genitalia.
- Clitoromegaly.
- Micropenis.
- Labio scrotal fold/fusion.
- Bifid scrotum.

M/C cause → in XY : Partial Androgen Insensitivity Syndrome.
→ in XX : Congenital Adrenal Hyperplasia (CAH) → medical emergency

----- Active space -----

Protocol for management of ambiguous genitalia :



Rx : Initiated irrespective of status of investigations.

HOMOLOGOUS GLANDS

Organs with same embryological origin.

Males	Females
Prostate gland	Skene gland/Paraurethral gland
Bulbourethral gland/Cowper's gland • Location : Deep perineal pouch.	Bartholin gland • Location : Superficial perineal pouch . • Origin in : Urogenital Sinus.

Note : Differentiation in male & female → From 12-14 weeks of intrauterine life.
(Sex determination by USG)

Pseudohermaphrodite :

	Male	Female
Karyotype	46 XY	46 XX
External genitalia	Female	Ambiguous
Cause	Complete Androgen Insensitivity Syndrome	CAH

----- Active space -----

DISORDERS OF SEXUAL DEVELOPMENT (46XX)

Disorders of sexual development in 46XX

Abnormal development of gonads :

1. Ovotestes
2. Turner's syndrome
   - 45 XO (m/c) → Streak ovary
   - Mosaic → 45 XO : Streak ovaries + 46 XY : Dysgenetic testes

Ambiguous genitalia :

1. Congenital adrenal hyperplasia m/c cause
2. ↑ maternal androgens :
   - Androgen secreting tumor
   - Luteoma
   - Androgenic drugs
3. Placental aromatase deficiency

[maternal androgen —aromatase→ Estrogen]

Ovotestes :

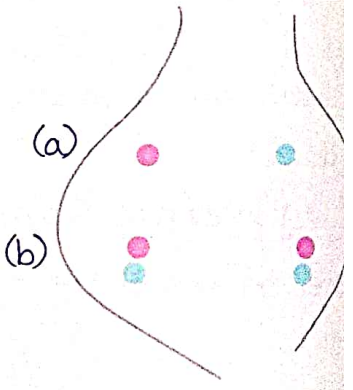
Old term : True hermaphrodite.

Karyotype : 46 XX > 46 XY.

Pattern :

a. 1 ovary & 1 testes on each side (m/c).

b. Ovarian + testicular tissue on both sides (L/c).



Internal genitalia : Both male + female internal genitalia.

On the side of ovary → Female internal genitalia.

On the side of testes → Male internal genitalia.

External genitalia :

- Ambiguous resembling male external genitalia d/t testosterone from adrenals.
- m/c presumed as male at birth.

At puberty → 3/4th cases : Gynecomastia (+) (d/t estrogen)
→ 1/4th cases : Menstruation (+)

Pathophysiology of CAH

00:12:05

m/c cause of → Ambiguous genitalia in females.
→ Disorder of sexual development (DSD) in 46XX.

Previous term : Female pseudohermaphroditism.

Adrenal gland physiology :

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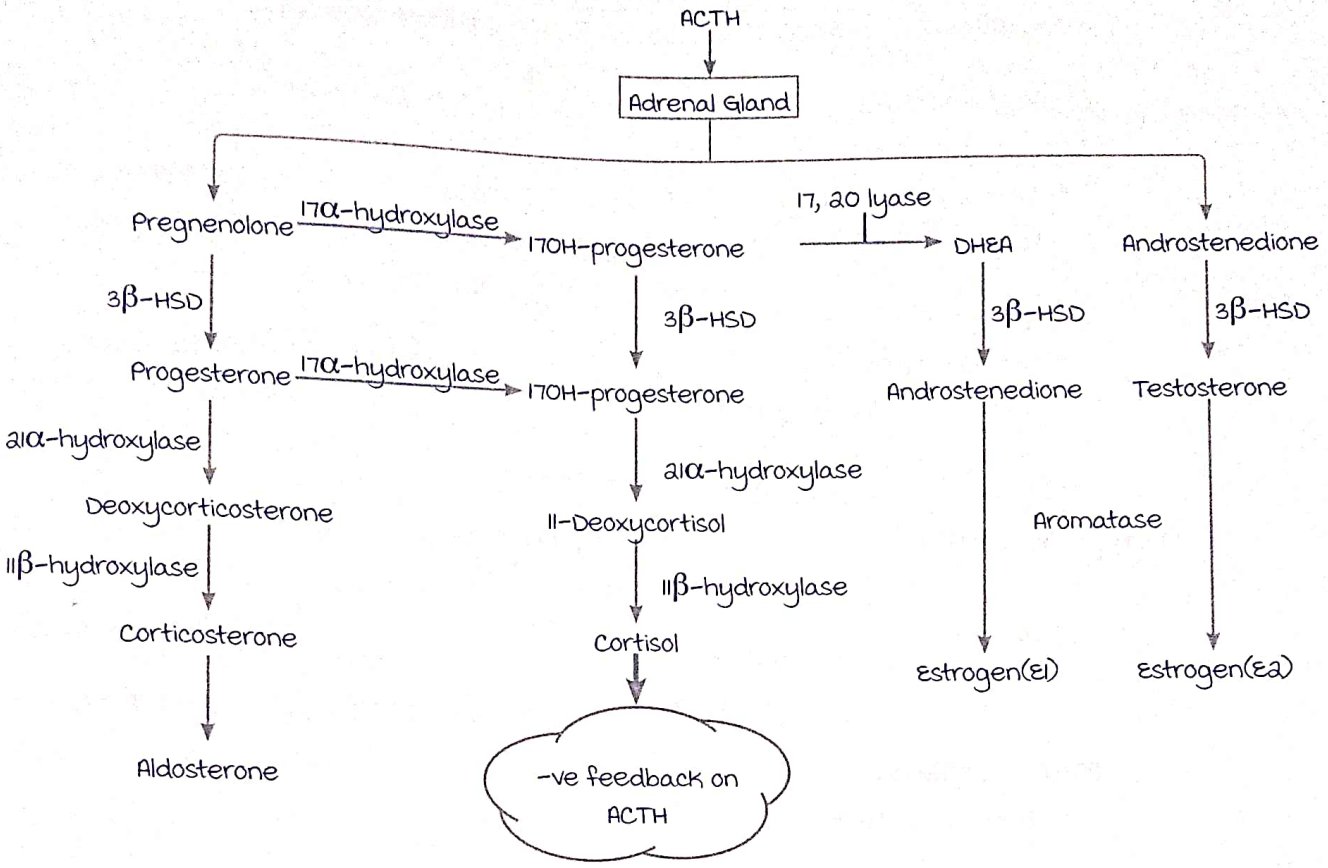

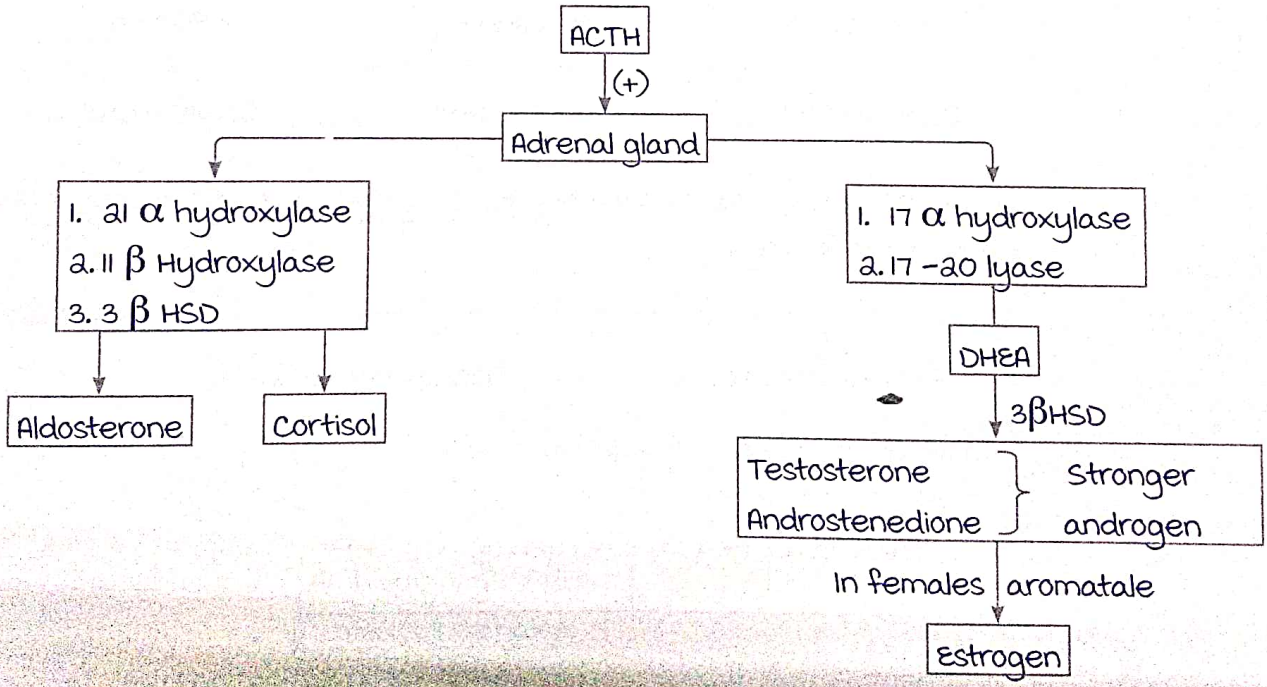

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21 α Hydroxylase & 11β Hydroxylase Deficiency

m/c deficiency in CAH : 21 α hydroxylase > 11 β hydroxylase.

Clinical features :

D/t ↓ aldosterone :

- Salt water wasting (dehydration).
- Hyponatremia.
- Hyperkalemia.
- metabolic acidosis.
- Hypotension.

D/t ↓ cortisol :

- Hypoglycemia.
- D/t loss of negative feedback :

↑ ACTH

- 17 OH progesterone ↑
- ↑ Androgens
  - Males : Normal external genitalia
  - Females : Ambiguous genitalia

Management :

1. Screening test : 17-OH progesterone
   - <200 → Rule out CAH
   - 200 – 800 ng → Diagnostic test : ACTH stimulation test
   - ≥ 800 ng → Confirm CAH
2. Treatment started without waiting for Karyotyping results based on physical examination findings
   - Gonads palpable
   - Gonads not palpable
3. Ambiguous genitalia + No palpable gonads → Treated as medical emergency

management

1. DOC : Glucocorticoids
2. mineralocorticoids
3. Correct electrolyte imbalance
4. Correction surgery for clitoromegaly later

Send for investigation later :

1. 17 OH progesterone
2. Serum electrolytes
3. Glucose
4. Karyotyping/FISH

management of pregnant female with h/o previous child with CAH :

DOC : Glucocorticoids - Dexamethasone.

↓

Started at 4-5 wks of pregnancy (maximum delay 9 wks)

↓

Chorionic villi sampling (sex determination + gene testing) at 10-11 wks

- male fetus/unaffected female fetus → ↓ Discontinue dexamethasone
- Affected female fetus → ↓ Continue dexamethasone

----- Active space -----

Note :

CAH : X-linked recessive disorder

Othe Causes for DSD

00:44:51

17 α HYDROXYLASE/17-20 LYASE DEFICIENCY

- Leads to DSD in XY individuals

↓ DHEA
↓ Androstenedione
↓ Testosterone

- Female fetus → ↓ Normal external genitalia
- male fetus → ↓ Ambiguous genitalia

3βHSD DEFICIENCY

- ↓ Aldosterone
- ↓ Cortisol
- ↓ Testosterone

↑ DHEA (weak androgen)

↓

Ambiguous genitalia males & females

- Male fetus : D/t ↓ Testosterone
- Female fetus : D/t ↑ DHEA

----- Active space -----

P450 OR DEFICIENCY

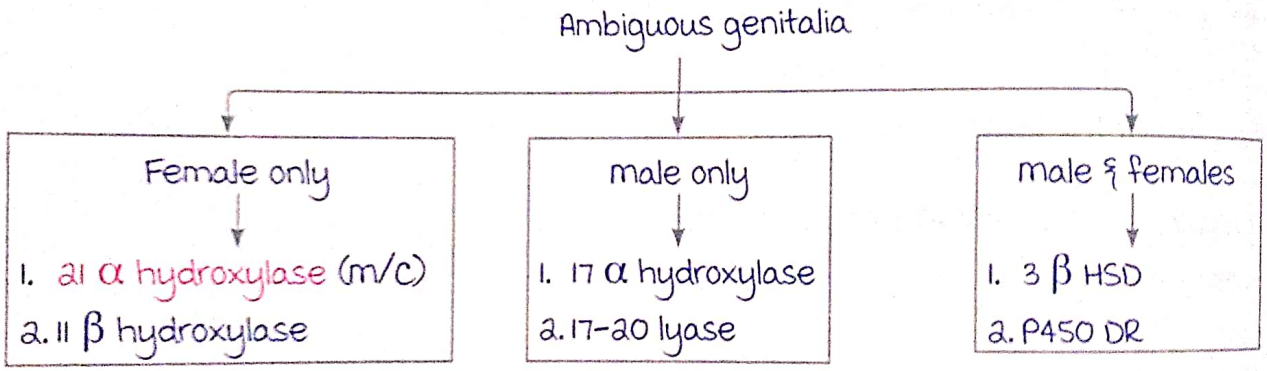
Newly diagnosed form of CAH.

Deficiency of P450 oxidoreductase enzyme (Chromosome 7).

P450 : Electron donor for many enzymes.

Affects both male & female fetuses : Ambiguous genitalia + Skeletal abnormalities (Eg. : craniosynostosis) } Antley Bixler syndrome

SUMMARY



DISORDERS OF SEXUAL DEVELOPMENT (46XY)

----- Active space -----

46 XY DSD (Disorders of Sexual Development)

Female external genitalia :

1. Complete androgen insensitivity syndrome (AIS) : m/c.
2. Swyer's syndrome.
3. 5α Reductase deficiency.
4. LH Receptor defect.

Ambigious genitalia :

1. Partial AIS.
2. Ovotestis.
3. Congenital Adrenal Hyperplasia (CAH) :
   - 17-20 hydroxylase deficiency.
   - 17 α lyase deficiency.
   - 3 β HSD deficiency.
   - P450 OR deficiency.

Characteristics :

- Considered as female child at birth :
  - Except : Ovotestis.
- Complaints of :
  1. 1° Amenorrhea at puberty :
     - Abdominal gonads : Palpable on physical examination.
  2. B/l Inguinal hernia ⟶ Undescended testis :
     - ↑ Risk of malignancy.
     - Rx : Gonadectomy.
     - m/c tumor associated with undescended testis : Gonadoblastoma.
     - m/c malignancy associated with undescended testis : Dysgerminoma.

- Female with 1° Amenorrhea + C/o Inguinal hernia : Suspect XY genotype.

Androgen Insensitivity Syndrome (AIS)

00:06:33

Location :

- Androgen Receptor : Long arm of X chromosome.
- SRY gene : Short arm of Y chromosome.

COMPLETE AIS

- X linked recessive disorder.
- mutation in androgen receptor ⟶ males are completely resistant to androgen, Testosterone & Dihydrotestosterone (DHT).

----- Active space -----

46 XY
SRY gene : N
↓
Testis : N

- Sertoli cells : N, FSH (+)
  - Antimullerian hormone.
    - ↓
    - Regression of mullerian duct.
      - ↓
      - No fallopian tubes, uterus, cervix, upper vagina.
  - Inhibin B.
    - ↓
    - -ve feedback on FSH.
      - ↓
      - FSH : N
      - Resistance to testosterone
        - Wolffian duct does not grow.
          - ↓
          - Wolffian duct regresses.
            - ↓
            - Failure of formation of male internal genitalia : No internal genital organs.
        - Failure of scrotal descent.
          - ↓
          - Abdominal testis.
            - ↓
            - ↑ Chance of malignancy B/L inguinal hernia.
        - Testosterone
          - ↓ 5α reductase
          - DHT
            - ↓
            - Resistant to DHT.
              - ↓
              - Failure of masculanisation of external genitalia.
                - ↓
                - Female external genitalia.
                - ↓
                - Female at birth.
- Leydig cells : N.
  - ↓
  - Testosterone : N but Resistant to testosterone.

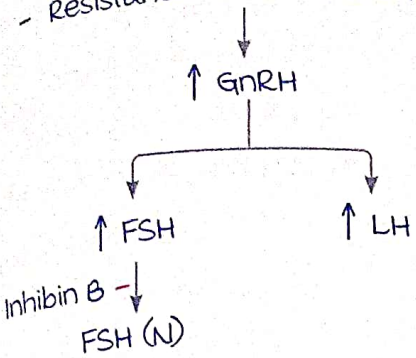
At puberty :

Testis → Testosterone
↓
Resistant

- Failure of 2° sexual development.
- Testosterone
  - ↓ Aromatase
  - Estrogen (Not resistant).
    - ↓
    - Breast development (Proper) : Tanner stage 4 or 5.
    - • may have blind ending vagina.
- Scanty pubic hair & axillary hair.
  - ↓
  - Tanner stage 1 or 2.

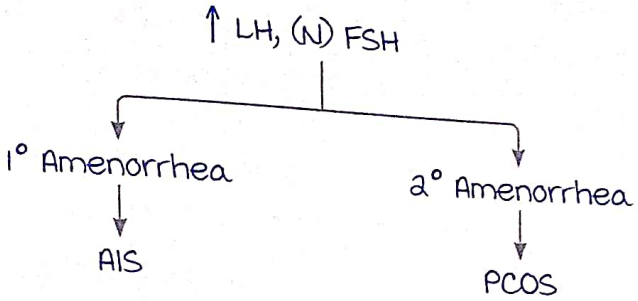
Hormonal profile :

Testosterone : -ve feedback on GnRH
- Resistance to testosterone



Inference :
FSH : N
LH : ↑
Testosterone : ↑ (Compared to females)

Note :



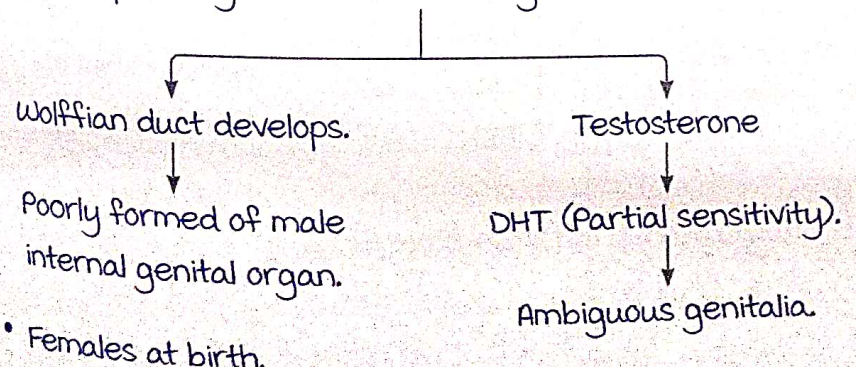
management :

- Continue identity of female.
- Gonadectomy of intraabdominal testis : After breast development completion 14 to 16 yrs.
- Estrogen replacement therapy : Progesterone : Not supplemented d/t absent uterus.
- Vaginoplasty :
  - Just before/after marriage.
  - Techniques :
    1. McIndoe vaginoplasty.
    2. Veicheitt technique : Laparoscopic.

Note : Complete AIS ♀ cannot have biological child.

PARTIAL AIS

Males are partially sensitive to androgens.



- Females at birth.

----- Active space -----

Features :

- 1° Amenorrhea d/t absent uterus.
- Breast : Tanner 4/5.
- Pubic & axillary hair :STanner 1 & 2.
- Clitoromegaly/Labioscrotal fusion.
- m/c cause of ambiguous genitalia in 46 XY.
- m/c cause of male pseudohermaphrodite.

Management :

- Same as AIS.
- Gonadectomy as soon as possible.
  - Testis → Testosterone → Ambiguous genitalia.
    → Virilisation at puberty.

Swyer's Syndrome

00:29:41

46 XY

↓

Gonads : Testis Dysgenetic →

- Causes :
  1. Idiopathic.
  2. Point mutation of SRY gene.
- Features :
  - ↑ Chances of malignancy.
  - Intra abdominal testis.

Sertoli cells (Dysgenetic).

- Absent AMH. → müllerian duct formation. → Female internal genitalia.
- Absent inhibin B. → ↑ FSH

Leydig cells (Dysgenetic). → Absent testosterone.

- Failure of Wolffian duct development : male internal genitalia absent.
- Absent -ve feedback on GnRH → ↑ GnRH, ↑ LH, ↑ FSH
- Absent DHT. → Female external genitalia. → Female child at birth.

----- Active space -----

At puberty :

No testosterone

- Failure of female 2° sexual characteristics formation : Absent breast.
- Failure of male 2° sexual characteristics formation.
- Pubic hair & axillary hair : Scanty.

Presentation :

- Delayed puberty.
- 1° Amenorrhea.
- B/L Inguinal hernia.

Management :

- Continue female identity.
- Gonadectomy : As soon as diagnosed.
- Estrogen supplementation :
  - For 1 year → Breast development.
  - Progesterone supplementation added : Lifelong.
- Vaginoplasty.

5α Reductase Deficiency

00:38:28

46 xy

Gonads : Testis (N)

- Sertoli cells (N) → AMH → Mullerian duct regress
- Leydig cells (N) → Testosterone → Wolffian duct development → male internal genital organ

Defect : Testosterone —5α reductase→ DHT →

Female external genital organs

Female at birth

----- Active space -----

At puberty :

Testis (N)
↓
Testosterone (N)
↓
Male 2° sexual characteristics ⟶ Virilisation :

- Clitoromegaly at puberty
- ↑ muscle mass
- Deepening of voice

- Absent breast development.
- Absent pubic or axillary hair : D/t absent DHT.
- 46 XY + Clitoromegaly at puberty + Testosterone level corresponding to females. ⟶ 1. Partial AIS / 2. 5α Reductase deficiency

Note :

In AIS : Testosterone ⟶ Estrogen.

Partial AIS vs 5α Reductase deficiency :

	Partial AIS	5α reductase deficiency
Time of clitoromegaly	Since birth	Puberty
Breast	Tanner 4/5	Absent
Internal genitalia	Poorly formed male genitalia	Fully formed male genitalia

PRIMARY AMENORRHEA

----- Active space -----

Features of Primary Amenorrhea

00:01:06

DEFINITION

- Absence of menstruation by 13 years + absence of breast budding (Thelarche).
- Absence of menstruation by 15 years + presence of breast budding.
- Thelarche $\xrightarrow{\geq 3 \text{ years}}$ Not menstruated

Note :

- Tanner Stages :
  - Breast budding : Stage 2.
  - Hair on mons pubis : Stage 3.

CAUSES

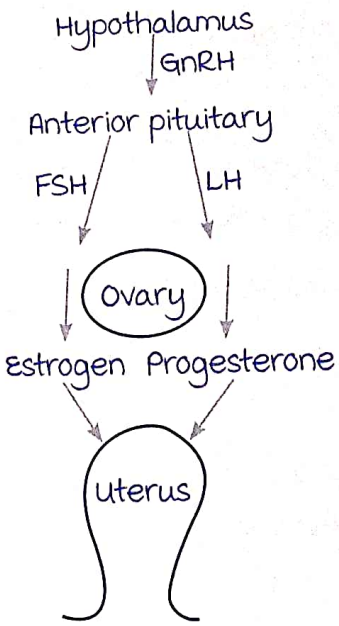
	Overall	India
m/c cause	Gonadal dysgenesis (Turner's syndrome > Swyer syndrome)	Mullerian agenesis
2nd m/c cause	Mullerian agenesis	Gonadal dysgenesis (Swyer's > Turner's)

Organic Causes :

Organ	Cause	Hormones
Hypothalamus	Kallman syndrome	↓ LH, ↓FSH, ↓E : Hypogonadotropic hypogonadism
Pituitary	Craniopharyngioma	
Ovary	• Gonadal dysgenesis : – 45XO (Turner's syndrome) : m/c – 46XY (Swyer syndrome) – 46XX (Pure gonadal dysgenesis)	↑ LH, ↑ FSH, ↓E (Testosterone ↓) : Hypergonadotropic hypogonadism
Uterus	Mullerian agenesis	LH, FSH, E : Normal
	Androgen Insensitivity Syndrome (AIS)	LH↑, FSH normal Testosterone : male levels

----- Active space -----

Uterus(−)	Uterus(+)
• mullerian agenesis (46XX) • AIS (46XY)	• Kallman syndrome • Gonadal dysgenesis • Cryptomennorhea



WORK UP OF PRIMARY AMENORRHEA

1. Physical Examination → Breast, Pubic hair
2. UPT : Especially if adequate breast development
3. USG : Know the status of uterus (Per rectal examination)
4. FSH, LH, E
5. Karyotyping : Best Ix

Conditions in the Presence of Uterus

00:18:41

1. CRYPTOMENORRHEA

Pathogenesis :

Normal menstruation; menstrual blood —D/t outflow tract obstruction→ Fails to come out

Causes :

1. Imperforate hymen : m/c cause.
2. Transverse vaginal septum (TVS).
3. Vaginal atresia /agenesis.

Findings :
- Karyotype : 46 XX.
- Gonads : Ovary.
- Genital organs : ♀, uterus (+).
- Pubic/Axillary hair : Normal.
- P/R examination : Blood collected in uterus → Hematometra → Enlarged Uterus.
- Breast : Normal.
- LH : Normal.
- FSH : Normal.
- E : Normal.

----- Active space -----

Presentation :
- Cyclical pain every month
- Primary amenorrhea
- +/- urinary retention.

→ Outflow tract obstruction (+)

Blood collection :
- In vagina : Hematocolpos.
- In uterus : Hematometra
  - +/- Supra pubic bulge

Retrograde menstruation → Endometriosis → ↑ dysmenorrhea

Examination :
- P/R: uterus (+) and Enlarged.
- Local examination.

- Nothing given → Imperforate hymen
- Tensed bulging hymen/ Cough Impulse (+) → Imperforate hymen
- Normal :
  - Hymen not bulging
  - Cough impulse (-)
  - → TVS
- Vaginal opening (-) → Vaginal agenesis

Investigations :

IOC : MRI.
- Differentiate between causes.
- Thickness/location of septa in TVS.

Management :

Imperforate hymen : Cruciate incision on hymen.

----- Active space ----- Other Conditions :

1. Gonadal dysgenesis	Similarities	2. Kallman syndrome
↓ Gonadal function ← D/t ← ↓ Estrogen → D/t →		↓ GnRH
45XO: Turner's syndrome (m/c)	- Breast development (-)	
46XY: Swyer syndrome	- Uterus : Hypoplastic (Infantile uterus)	
	• Scanty pubic/axillaty hair	
ε ↓, FSH ↑, LH ↑		LH ↓, FSH ↓, ε ↓

2. GONADAL DYSGENESIS

a) Turner's Syndrome :

Features :

- Karyotype :
  - m/c : 45 XO
  - Turner mosaic : 45XO/46XY (Rare)
- Gonads : Streak ovaries → ↓ estrogen ← Streak ovary + dysgenetic testis
  - Non-malignant : Should not be removed
  - ↑ Chance of malignancy : Should be removed
- Stature : Short d/t mutation of SHOX gene.

Other features :

- Webbing of neck.
- Low posterior hair line.
- Shield shaped chest.
- Widely spaced nipples.
- SNHL (Sensorineural hearing loss).
- Autoimmune diseases : Diabetes/Hashimoto thyroiditis/Celiac disease.
- Cubitus valgus.
- Cardiac disease : Bicuspid aortic valve > Coarctation of aorta.

Investigations :

- Karyotyping : Rule out mosaic.
- ECHO (Heart disease).
- Renal USG (A/w horse shoe kidney).
- Thyroid profile (Hashimoto thyroiditis).

- Fasting glucose, lipid profile (DM).
- Audiometry (SNHL).
- Anti endomyseal antibodies (Celiac disease).

----- Active space -----

Management :

- Growth Hormone : Before 8 years of age.
- Estrogen : Breast development X 1 year
  ↓
  E+P : Because uterus (+)
- Pregnancy : Relative c/i d/t cardiac abnormalities.

b) Swyer Syndrome :

Presentation :

- ♀ c/o primary amenorrheaa.
- Uterus (+), Breast development(–).
- Height : Normal.
- FSH, LH : ↑.
- Gonads : Dysgenetic testis.

Turner's vs Swyer Syndrome :

	Turner's Syndrome	Swyer Syndrome
Karyotype	45XO	46XY
Gonads	Streak ovaries	Dysgenetic testis ⟶ B/l inguinal hernia
Nature	Absent	Normal
Additional features	+	–

3. KALLMAN SYNDROME

Findings :

- Karyotype : 46XX.
- Gonads : Ovary (Normal function).
- Pathology : Hypothalamic failure (↓ GnRH ⟶ ↓LH, ↓FSH, ↓E).
- m/c gene affected : KAL-1 gene.
- Genital organs : ♀, hypoplastic uterus d/t ↓E.
- Breast : Not well developed.
- Scanty pubic/axillary hair.
- Hormones : FSH, LH ↓.
- Stature : Normal.
- A/w : Anosmia.

----- Active space -----

Presentation :

- Delayed puberty.
- 1° amenorrhea.
- Infertility.

Management :

- Pulsatile GnRH.

Conditions in the Absence of Uterus

00:45:45

	mullerian agenesis	AIS
Karyotype	46XX	46XY
Gonads	Ovaries (Normal as they develop from genital ridge)	Testis (Normal)
Internal genitalia	Absent (Both mullerian ducts (MD) (-) → Fallopian tubes, uterus, cervix, upper vagina absent)	Absent
External genitalia	♀	♀
Breast development	Estrogen : Normal → Breast development : Normal	Tanner stage 4/5
Pubic & axillary hair	Androgens : Normal hair distribution	Scanty
Associated problem	• Mayer-Rokitansky-Kuster-Hauser (MRKH) syndrome • B/L MD (-) • Renal anomalies : Agenesis/Horse shoe kidney • Skeletal anomalies : Hemivertebra	Unresponsive to testosterone
Testosterone	Normal : Equivalent to female	↑ (Equivalent to ♂ levels)
LH	Normal	↑
FSH	Normal	Normal
Produce biological children	Yes : IVF with oocyte + sperm → Surrogacy	No
Investigations	In all patients : 1. IVF (Intra venous pyelography) : Renal anomalies 2. Skeletal X-ray : Vertebral anomalies	-
management	Vaginoplasty (For coital problems) : - Time : Just before/after marriage - Technique → McIndoe technique / Veicheitt technique	• Rx as ♀ • Gonadectomy • Vaginoplasty • Estrogen replacement therapy

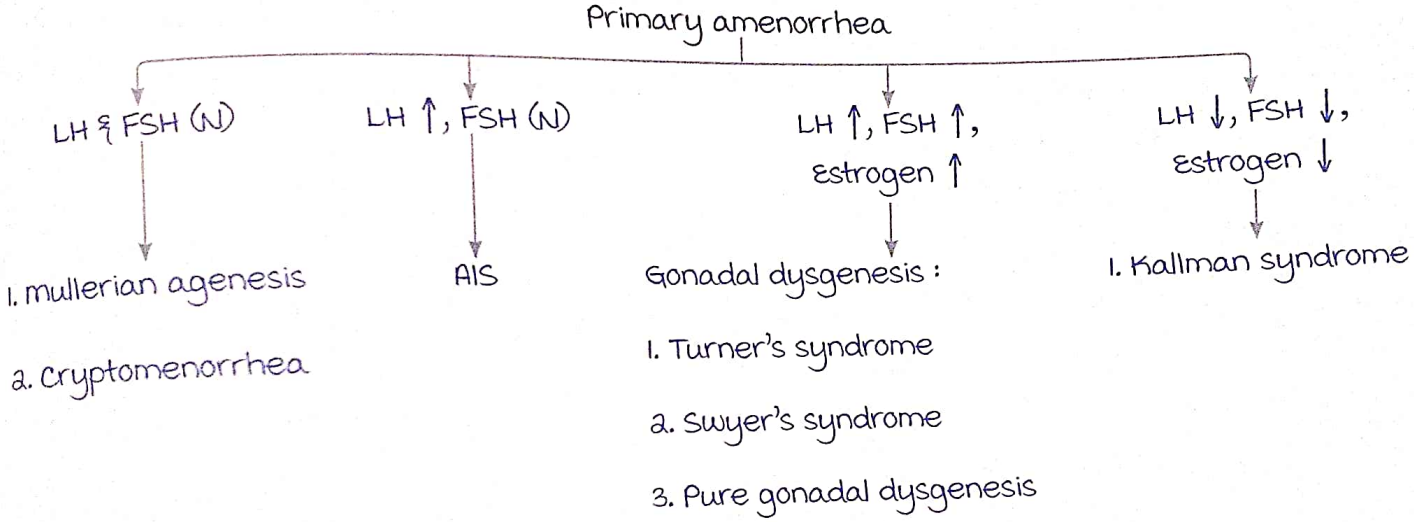
MURCS Association :
Mullerian agenesis
Renal anomalies
Cervical Somite

----- Active space -----

Features :

- Both fallopian tubes absent
  - Sometimes distal end (+)
- Upper vagina (–), lower vagina (+): Derived from urogenital sinus
  - Sometimes, entire vagina(–) : Vaginal agenesis

SUMMARY



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VAGINITIS AND CERVICITIS

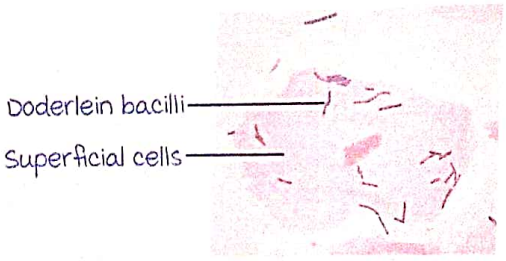
Vaginal defense mechanism :

Vaginal flora : Polymicrobial.

- Lactobacilli / Doderlein bacilli (Predominant) :
  - Gram +ve bacilli.
  - Appears at puberty.
  - Disappears at menopause.
  - Glycogen in stratified epithelium → Lactic acid + H_2O_2
    → Acidic pH → Defense mechanism.
  - Estrogen : Responsible for presence of glycogen.
  - Candida : Survives in acidic pH.

Pap smear :

Screening for cancer.



Normal pap smear

Leucorrhoea :

- Physiological white discharge.
- Increased :
  - During ovulation.
  - Before menstruation.

Characteristics :

- Colourless.
- Odourless.
- Consistency :
  - 1st half of cycle : Thin watery d/t estrogen.
  - and half of cycle : Thick d/t progesterone.
- No pruritis/dysuria.

D/d :

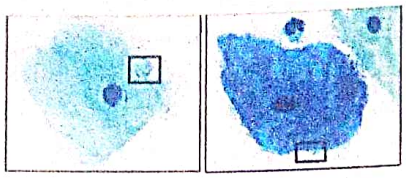
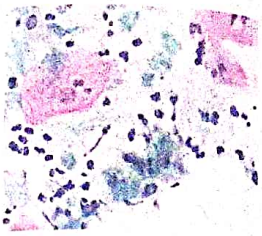
- Bacterial vaginosis discharge :
  - Colourless.
  - No pruritis.
  - Pungent/fishy odour.

00:06:53 ----- Active space -----

Vaginitis

	Bacterial vaginosis	Candidiasis	Trichomonas vaginitis
Organism	• Gardnerella vaginalis • mycoplasma • mobiluncus • Alteration of vaginal flora • m/c vaginitis in female	Candida albicans (Fungus)	Trichomonas vaginalis : Flagellated protozoan
Predisposing factors	Alteration of vaginal pH (Alkaline)	• Pregnancy – m/c vaginitis in pregnant female • Immunocompromised • Antibiotics, steroids, OCP consumption	Coexist with bacterial vaginosis
Discharge	• Greyish white / dirty white • Foul smell	• Scanty • Curdy white / cottage cheese • Not main complaint	• Frothy • Yellowish green color • Foul smell
pH	≥ 4.5	< 4.5	≥ 4.5
Pruritis	X (No inflammation)	Intense pruritis → Excoriation : main complaint	Intense pruritis
Dysuria	X	Splash dysuria	✓
Polymorphonucleocytes (PMNs) / epithelial cells	< 1	≥ 2	≥ 2
STD	X	Not usually	✓

----- Active space -----

	Bacterial vaginosis	Candidiasis	Trichomonas vaginitis
IOC : Saline microscopy	Epithelial cells with bacteria adhered : Clue cell  Normal vaginal flora — Clue cells	• Pseudohyphae • Spores of candida Candidiasis	• motility + (or) • Tennis racket appearance  Pear shaped organism ≈ Size of nucleus of cells.
Gold standard test	• Gram staining – Nugent score : Score 7-10	Culture on Sabouraud media	Culture
Named test	• Whiff test : Add 10% KOH to discharge → fishy / amine like odour • Amsel's criteria : 1. Greyish White discharge 2. pH ≥ 4.5 3. Clue cells ≥ 20% of epithelial cells 4. Whiff test : +ve If 3/4 positive → Diagnostic	Whiff test -ve	• Whiff test: +ve • P/s examination : Strawberry cervix/ Angry looking vagina
DOC : Non-pregnant	• metronidazole : Oral 500 mg BD x 7days • Clindamycin	Oral fluconazole : 150 mg stat	• metronidazole : – 500 mg BD x 7 days (or) – 2 gram stat